

Amended Draft Package Insert

LYCOMINE SYRUP

Each 5ml contains: -

Lysine HCl	308.5 mg
Cyanocobalamin	3 mcg

Pharmacology:

Lysine HCl is required for growth and bone development in children, assists in calcium absorption and maintaining the correct nitrogen balance in the body and maintaining lean body mass. Furthermore it is needed to produce antibodies, hormones, enzymes, collagen formation as well as repair of tissue. Since it helps with the building of muscle protein, it is useful for patients recovering from injuries and recovery after operations, and there might be use in lysine to help maintain healthy blood vessels. Cyanocobalamin (Vitamin B12), a water-soluble vitamin, occurs in the body mainly as methylcobalamin (mecobalamin) and as adenosylcobalamin (cobamamide) and hydroxocobalamin. Mecobalamin and cobamamide act as coenzymes in nucleic acid synthesis. Mecobalamin is also closely involved with folic acid in several important metabolic pathways.

Pharmacokinetics:

Following ingestion, lysine is absorbed from the lumen of the small intestine into the enterocytes by an active transport process. Some metabolism of lysine takes place within the enterocytes. That which is not metabolized is transported to the liver via the portal circulation. In the liver, lysine, along with other amino acids, participates in protein biosynthesis. Some is metabolized to L-alpha-aminoadipic acid semialdehyde, which is further metabolized to acetoacetyl-CoA. The intermediate in this pathway is saccharopine. Lysine does not participate in transamination. It is the exception to the general rule that the first step in catabolism of an amino acid is the removal of its alpha-amino group by transamination to form the respective alpha-keto acid. Lysine is both a glycogenic and a ketogenic amino acid. It can participate in the formation of D-glucose and glycogen, as well as lipids. It can also participate in the production of energy.

Vitamin B12 substances is bound to intrinsic factor and are then actively absorbed from the gastro-intestinal tract. Absorption from the gastro-intestinal tract can also occur by passive diffusion; little of the vitamin present in diets is absorbed in this manner although the process becomes increasingly important with larger amounts such as those used therapeutically. Vitamin B12 is extensively bound to specific plasma proteins called transcobalamins; transcobalamin II appears to be involved in the rapid transport of the cobalamins to tissues. It is stored in the liver, excreted in the bile, and undergoes extensive enterohepatic recycling. Part of an administered dose is excreted in the urine, most of it in the first 8 hours; urinary excretion, however, accounts for only a small fraction in the reduction of total body stores acquired by dietary means.

Indications:

As a dietary supplement.

Contraindications:

It is contraindicated in patients who are sensitive to any of the ingredients.

Side effects/Adverse reactions:

No reports of side effects, though occasional hypersensitivity reactions have occurred.

Warnings and Precautions:

Severe metabolic acidosis have been reported with Lysine HCl. Lysine HCl should be administered with caution to patients with hyperchloraemic acidosis or impaired renal function.

The absorption of cyanocobalamin is reduced by metformin, potassium citrate and/or calcium gluconate.

Ascorbic acid destroys cyanocobalamin in-vitro. The destructive effect is affected by the concentration of both exogenous and food source of cyanocobalamin.

Pregnancy and Lactation:

Pregnant women and nursing mothers should only consider using supplemental lysine if their diets are low in this amino acid. They should avoid supplemental lysine for other reasons.

Recommended Dose:

Oral administration.

Children (below 4 years of age) : ½ to 1 teaspoonful daily.

Children (above 4 years of age) and adults : 2 teaspoonfuls daily.

Drug Interactions:

Concomitant use of calcium supplements and lysine may increase calcium absorption. This is based on a very preliminary study that needs follow-up.

Symptoms and Treatment for overdose and antidote(s):

There are no reports on the symptoms / effects of overdose.

Therapeutic or accidental overdose resulting in adverse reactions or fatalities is quite unlikely, considering its recommended/therapeutic dose and it being very well tolerated.

Storage conditions: Store at or below 25°C. Protect from light.

Pack size: A bottle of 60ml, 100ml and 120ml.

Pack size (Export only) : A bottle of 1 litre.

Shelf life: 2 years.

Description: A clear, orange coloured syrup with orange flavour.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.

Manufacturer:

Sunward Pharmaceutical Sdn. Bhd.

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